ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

MASTER OF SCIENCE IN DATA SCIENCE

CAS5821 MATHEMATICAL FOUNDATIONS OF DATA SCIENCE

FINAL EXAMINATION

DATE: 3RD JUNE 2025

TIME: 14:00 – 17:00 HRS

TIME ALLOWED : WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted

Question 1

(a) Explain five differences between Regression Analysis and Correlation Analysis. [5 Marks]

(b) The equation used to predict the total weight of a female athlete at a certain school is $\hat{Y}=-110+2.88x_1+2.04x_2$ where x_1 is the athlete's height (in inches) and x_2 is the female athlete's percent body fat measured as x_2%. Use the multiple regression equation to predict the total body weight for a female athlete who is 72 inches tall and has 27% body fat. [10 Marks]

(c) The table below gives the wine consumption in Zambia in millions of litres (y) for the years 1963 to 1972 (x).

Years (x)	1963	1964	1965	1966	1967	1968	1969	1970	1971	1972
Cons. (y) L(m)	32.5	37.1	35.5	37.7	41.5	46.4	44.5	45.8	53.9	62.0

(i) Calculate the equation of the regression line of y on x. [15 Marks]

(ii) Calculate the Pearson Correlation Coefficient and interpret its meaning. [15 Marks]

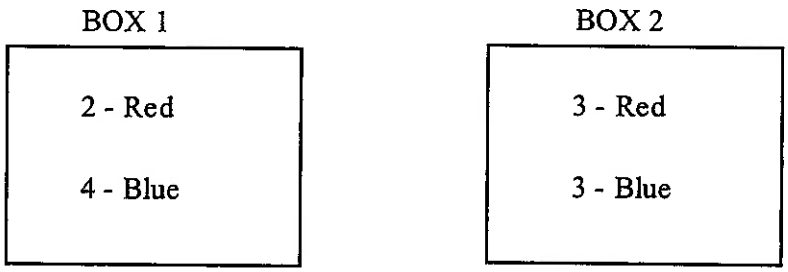
(iii) Is this linear regression model appropriate given also that the actual wine consumption in 1973 was 78.3million litres? Explain your answer. [5 Marks]

(Total: 50 marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

(a) Suppose you have two boxes labelled Box 1 and Box 2. In Box 1 there are 2 red chalks and 4 blue chalks. In Box 2 there are 3 red chalks and 3 blue chalks. The probability of picking any one of the two boxes is half. Within the boxes the probability of picking a red or a blue chalk is given below:



P(Box 1) = 1/2
P(Red|Box 1) = 2/6
P(Blue|Box 1) = 4/6

P(Box 2) = 1/2
P(Red|Box 2) = 3/6
P(Blue|Box 2) = 3/6

Let's assume that the two boxes are covered and you are asked to pick a chalk. Not knowing which box your hand is in, you pull out a red chalk. What is the probability that the red chalk came from Box 1? [10 Marks]

(b) Find $\int \frac{4}{5x-3} dx$ [5 Marks]

(c) Let $A = \begin{pmatrix} 1 & -1 & -2 \\ 0 & 2 & -1 \\ -1 & 3 & 1 \end{pmatrix}$. Find all the Eigen Values and Eigen Vectors of A. [10 Marks]

(25 marks)

Question 3

(a) An expert backgammon player is currently competing in two tournaments. He estimates the probability of winning the first tournament as 0.6 and the probability of winning the second tournament as 0.5. If he wins the first tournament, he estimates the probability of winning the second tournament increases to 0.65 because of an increase in confidence. What is the probability of:

(i) the player winning both tournaments? [5 Marks]

(ii) the player winning at least one tournament? [5 Marks]

(b) Let $A = \begin{pmatrix} 1 & -2 & 3 \\ 2 & -5 & 12 \\ 0 & 2 & -10 \end{pmatrix}$. Factorize as a product of two matrices L and U. [5 Marks]

(c) Find $\frac{dy}{dx}$ of the following:

(i) $y = x^3 + 3x^2 - 1$ (ii) $y = \sqrt{2x-1}$ [10 Marks]

(25 marks)

Question 4

An engineer is interested in studying the relationship between the temperature and resistance of vacuum transducer bobbins, which are used in the automobile industry. She believes that this relationship can be studied in a linear regression model, $Y = \beta_0 + \beta_1 x$ where x is temperature and y is the resistance. After collecting data, she used Minitab to estimate the regression model. Below is the partial Minitab output:

Regression Analysis: Resistance Versus Temperature					
Analysis of variance					
Source	DF	SS	MS	F	P
Regression	1	624.70	624.703	156.91	0.000
Error	22	87.59	?		
Total	?	712.29			

Coefficients				
Predictor	Coef	SE Coef	T	P
Constant	?	4.55	2.83	0.010
Temparature	?	0.06	12.60	0.000

(a) How many observed pairs of x and y did her data set have? [5 Marks]

(b) What is the expected value of resistance when the temperature is 70 degrees? [10 Marks]

(c) What is the coefficient of determination, r^2? [10 Marks]

(25 marks)

END OF EXAMINATION